

Beer Reviews EDA

This project explores the global beer market using a comprehensive dataset of 1.5 million reviews from Beer Advocate. We analyze ratings for aroma, taste, appearance, and palate to uncover what defines the world's top-rated beers.

Our goals:

- Identify popular beer styles and trends.
- Spotlight top breweries.
- Understand what factors influence high ratings.

Importing Libraries and reading data

```
In [31]: #Importing packages
import pandas as pd
import numpy as np
import statsmodels.api as sm
import seaborn as sns
import matplotlib.pyplot as plt
import zipfile
import sklearn as sk
import scipy.stats as ss
from scipy.stats import boxcox
from scipy.special import inv_boxcox
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder
from sklearn.model_selection import StratifiedShuffleSplit, GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier, StackingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import VotingClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from imblearn.over_sampling import SMOTE, RandomOverSampler

In [32]: #reading data
beer_data= pd.read_csv('beer_reviews.csv')
beer_data.head()
```

Out[32]:

	brewery_id	brewery_name	review_time	review_overall	review_aroma	review_appearance	review_profilename	beer_style	review_palate	review_taste	beer_name	beer_abv	beer_beerid
0	10325	Vecchio Birraio	1234817823	1.5	2.0	2.5	stcules	Hefeweizen	1.5	1.5	Sausa Weizen	5.0	47986
1	10325	Vecchio Birraio	1235915097	3.0	2.5	3.0	stcules	English Strong Ale	3.0	3.0	Red Moon	6.2	48213
2	10325	Vecchio Birraio	1235916604	3.0	2.5	3.0	stcules	Foreign / Export Stout	3.0	3.0	Black Horse Black Beer	6.5	48215
3	10325	Vecchio Birraio	1234725145	3.0	3.0	3.5	stcules	German Pilsener	2.5	3.0	Sausa Pils	5.0	47969
4	1075	Caldera Brewing Company	1293735206	4.0	4.5	4.0	johnmichaelsen	American Double / Imperial IPA	4.0	4.5	Cauldron DIPA	7.7	64883

Data Cleaning

Removing Duplicates

We drop the `review_time` column to detect and remove duplicate reviews by the same user for the same beer. This ensures our analysis is not biased by repeated ratings.

```
In [33]: #Checking for Duplicates
beer_data.drop(['review_time'], axis=1, inplace=True)
display(pd.DataFrame(beer_data[beer_data.duplicated()]))
print(f"Number of duplicates removed: {beer_data.duplicated().sum()}")
beer_data=beer_data.drop_duplicates(keep='first')
```

	brewery_id	brewery_name	review_overall	review_aroma	review_appearance	review_profile	beer_style	review_palate	review_taste	beer_name	beer_abv	beer_beerid
1639	1454	Broad Ripple Brew Pub	4.5	3.5	3.5	danielocean03	Märzen / Oktoberfest	4.0	3.5	Oktoberfest	5.50	13289
7506	1386	Harmon Restaurant & Brewery	4.5	4.0	4.0	barleywinefiend	American Double / Imperial IPA	4.5	4.5	Commencement Bay	NaN	66872
7577	1386	Harmon Restaurant & Brewery	4.5	4.5	4.5	barleywinefiend	American Stout	4.5	4.5	Stryker	NaN	73647
7655	1482	The Seven Barrel Brewery	4.0	3.5	3.5	akorsak	American Amber / Red Ale	4.0	3.5	Red 7	NaN	7046
10064	5049	Bath Ales Ltd	4.0	3.5	4.0	wl0307	English Bitter	4.0	4.0	Barnstormer	4.50	13084
...	...	...	...	...	...	...	...	...	...	...	...	...
1575063	364	Maritime Pacific Brewing Company	4.0	4.0	4.0	barleywinefiend	American Double / Imperial IPA	4.0	4.0	Imperial Pale Ale	7.50	957
1582040	429	Alaskan Brewing Co.	4.0	4.0	4.0	JayQue	American IPA	4.0	4.0	Alaskan IPA	6.20	22787
1584107	41	Brouwerij Corsendonk	4.0	4.5	4.0	goindownsouth	Dubbel	4.0	4.0	Corsendonk Pater / Abbey Brown Ale	7.50	138
1585815	3835	Drake's Brewing Co.	4.5	4.0	4.0	rayjay	Russian Imperial Stout	4.5	4.0	Drake's Drakonic Imperial Stout	8.75	12172
1586581	14359	The Defiant Brewing Company	5.0	4.5	4.5	EPseja	Irish Dry Stout	4.5	4.5	O'Defiant Stout	5.50	36388

774 rows × 12 columns

Number of duplicates removed: 774

```
In [34]: display(pd.DataFrame(beer_data[beer_data.duplicated()]))
```

brewery_id	brewery_name	review_overall	review_aroma	review_appearance	review_profilename	beer_style	review_palate	review_taste	beer_name	beer_abv	beer_beerid
------------	--------------	----------------	--------------	-------------------	--------------------	------------	---------------	--------------	-----------	----------	-------------

Duplicate entries Deleted

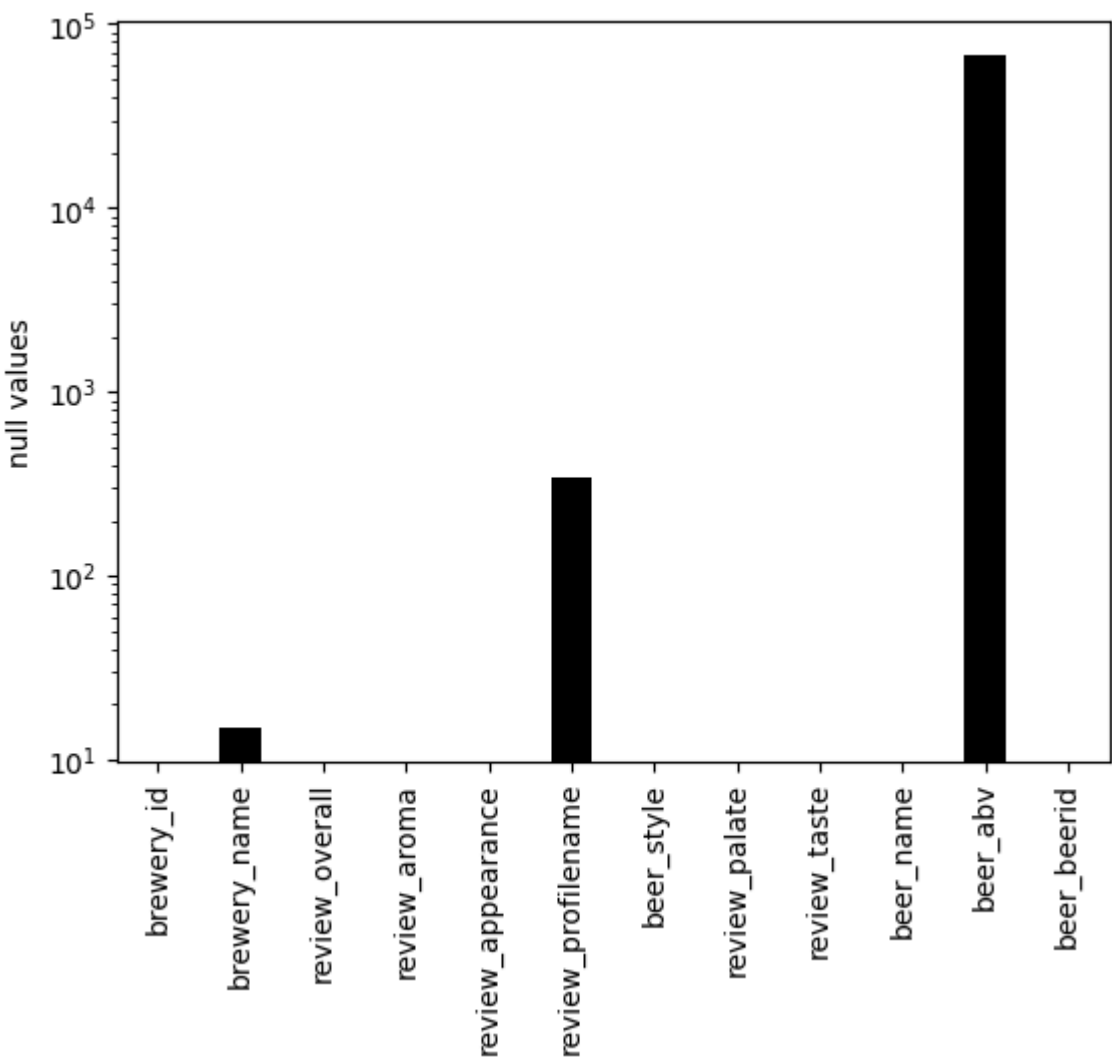
Null Values

```
In [35]: # Checking Nulls
beer_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1585840 entries, 0 to 1586613
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   brewery_id            1585840 non-null  int64
1   brewery_name          1585825 non-null  object
2   review_overall        1585840 non-null  float64
3   review_aroma          1585840 non-null  float64
4   review_appearance     1585840 non-null  float64
5   review_profilename    1585493 non-null  object
6   beer_style            1585840 non-null  object
7   review_palate         1585840 non-null  float64
8   review_taste          1585840 non-null  float64
9   beer_name             1585840 non-null  object
10  beer_abv              1518078 non-null  float64
11  beer_beerid           1585840 non-null  int64
dtypes: float64(6), int64(2), object(4)
memory usage: 157.3+ MB
```

```
In [36]: null_counts=beer_data.isnull().sum()
null_counts.plot(kind='bar',color='black')
plt.yscale('log')
plt.ylabel('null values')
```

Out[36]: Text(0, 0.5, 'null values')



Nulls found in : brewery_name and beer_abv

Lets start with nulls in brewery_name column

```
In [37]: display(beer_data[beer_data['brewery_name'].isnull()])
```

	brewery_id	brewery_name	review_overall	review_aroma	review_appearance	review_profile	beer_style	review_palate	review_taste	beer_name	beer_abv	beer_beerid
651565	1193	NaN	2.0	2.5	2.5	Knapp85	Vienna Lager	1.5	1.5	Engel Tyrolian Bräu WRONG BREWERY SEE SCHWABIS...	5.0	67503
659293	1193	NaN	4.0	4.5	3.5	dqrull	Bock	4.0	3.5	Engel Bock Dunkel WRONG BREWERY SEE CRAILSHEIMER	7.2	63658
659299	1193	NaN	3.5	3.0	3.0	dqrull	Dortmunder / Export Lager	4.0	4.0	Engel Gold WRONG BREWERY SEE CRAILSHEIMER	5.4	63215
659300	1193	NaN	3.5	4.0	3.5	dqrull	Munich Helles Lager	3.5	3.0	Engel Landbier WRONG BREWERY SEE CRAILSHEIMER	4.8	63557
659301	1193	NaN	3.5	4.0	4.0	dqrull	Keller Bier / Zwickel Bier	4.0	3.5	Engel Keller Hell WRONG BREWERY SEE CRAILSHEIMER	5.4	63256
659302	1193	NaN	3.5	4.0	3.0	dqrull	Vienna Lager	3.0	3.5	Engel Aloisius - WRONG BREWERY SEE CRAILSHEIMER	5.9	63459
659303	1193	NaN	3.0	3.0	3.0	Ochsenblut	Keller Bier / Zwickel Bier	2.0	3.0	Engel Keller Dunkel WRONG BREWERY SEE CRAILSH...	5.3	63324
659304	1193	NaN	4.5	4.0	4.0	Dentist666	Keller Bier / Zwickel Bier	4.0	4.5	Engel Keller Dunkel WRONG BREWERY SEE CRAILSH...	5.3	63324
659305	1193	NaN	3.5	3.5	4.0	dqrull	Keller Bier / Zwickel Bier	4.0	4.0	Engel Keller Dunkel WRONG BREWERY SEE CRAILSH...	5.3	63324
1391043	27	NaN	3.5	3.5	3.5	Jason	American Adjunct Lager	4.0	3.5	Hard Hat American Beer	3.8	60
1391049	27	NaN	2.5	2.0	2.0	BeerBob	American Malt Liquor	2.0	2.0	Side Pocket	8.5	3394
1391050	27	NaN	1.5	1.5	2.0	feloniousmonk	American Malt Liquor	1.5	1.5	Side Pocket	8.5	3394
1391051	27	NaN	4.5	4.5	3.5	ccrida	American IPA	4.5	4.5	Breakaway IPA	7.2	75135
1391052	27	NaN	3.5	4.0	3.5	Docer	American IPA	4.0	4.0	Breakaway IPA	7.2	75135
1391053	27	NaN	5.0	4.5	4.0	Docer	American Stout	4.5	4.5	Caboose Oatmeal Stout	7.0	75137

```
In [38]: #Replacing Nulls
beer_data.loc[beer_data['brewery_id'] ==1193,'brewery_name'] = 'CRAILSHEIMER'
beer_data.loc[beer_data['brewery_id'] ==1193,'beer_name']=beer_data.loc[beer_data['brewery_id'] ==1193,'beer_name'].str.replace('WRONG BREWERY SEE CRAILSHEIMER','',regex=True)
beer_data.loc[beer_data['brewery_id'] ==1193,'beer_name']=beer_data.loc[beer_data['brewery_id'] ==1193,'beer_name'].str.replace('WRONG BREWERY SEE SCHWABISCH GMUND','',regex=True)
beer_data['beer_name']=beer_data['beer_name'].str.strip()
beer_data.loc[beer_data['brewery_id'] ==27,'brewery_name'] = 'Unknown'
```

```
In [39]: #Reviews by style
review_by_style= beer_data.groupby(['beer_style'])['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()

/tmp/ipykernel_229101/1404892529.py:2: FutureWarning: Indexing with multiple keys (implicitly converted to a tuple of keys) will be deprecated, use a list instead.
review_by_style= beer_data.groupby(['beer_style'])['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()
```

beer_abv Nulls

- Replaced with the mean of beer_abv

```
In [40]: bd_null= beer_data.loc[beer_data['beer_abv'].isnull(),'beer_abv'].index
beer_data.beer_abv[bd_null]= review_by_style.beer_abv[beer_data.beer_style[bd_null]]
```

```
#Final null check
print(beer_data.isnull().sum())
```

/tmp/ipykernel_229101/721401938.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
beer_data.beer_abv[bd_null]= review_by_style.beer_abv[beer_data.beer_style[bd_null]]
```

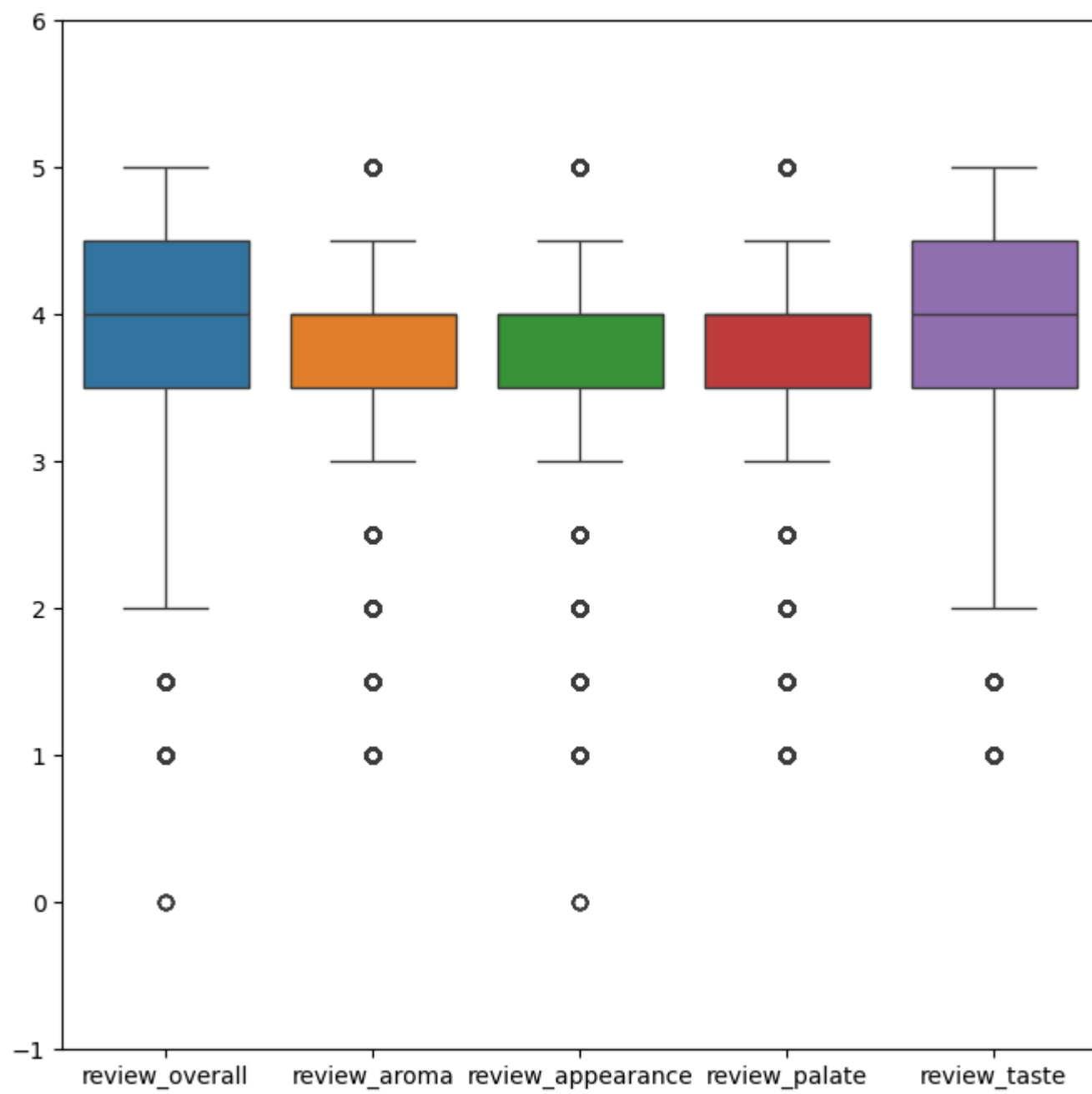
```
brewery_id      0
brewery_name    0
review_overall  0
review_aroma    0
review_appearance  0
review_profilename  347
beer_style      0
review_palate   0
review_taste    0
beer_name       0
beer_abv        0
beer_beerid     0
dtype: int64
```

```
In [41]: #Removing irrelevant columns
beer_data.drop(['beer_beerid', 'brewery_id', 'review_profilename'], axis=1, inplace=True)
```

Outliers

```
In [12]: #Outlier detection
plt.figure(figsize=(8,8))
sns.boxplot(data=beer_data.drop(columns=['beer_abv']))
plt.ylim(-1,6)
```

Out[12]: (-1.0, 6.0)

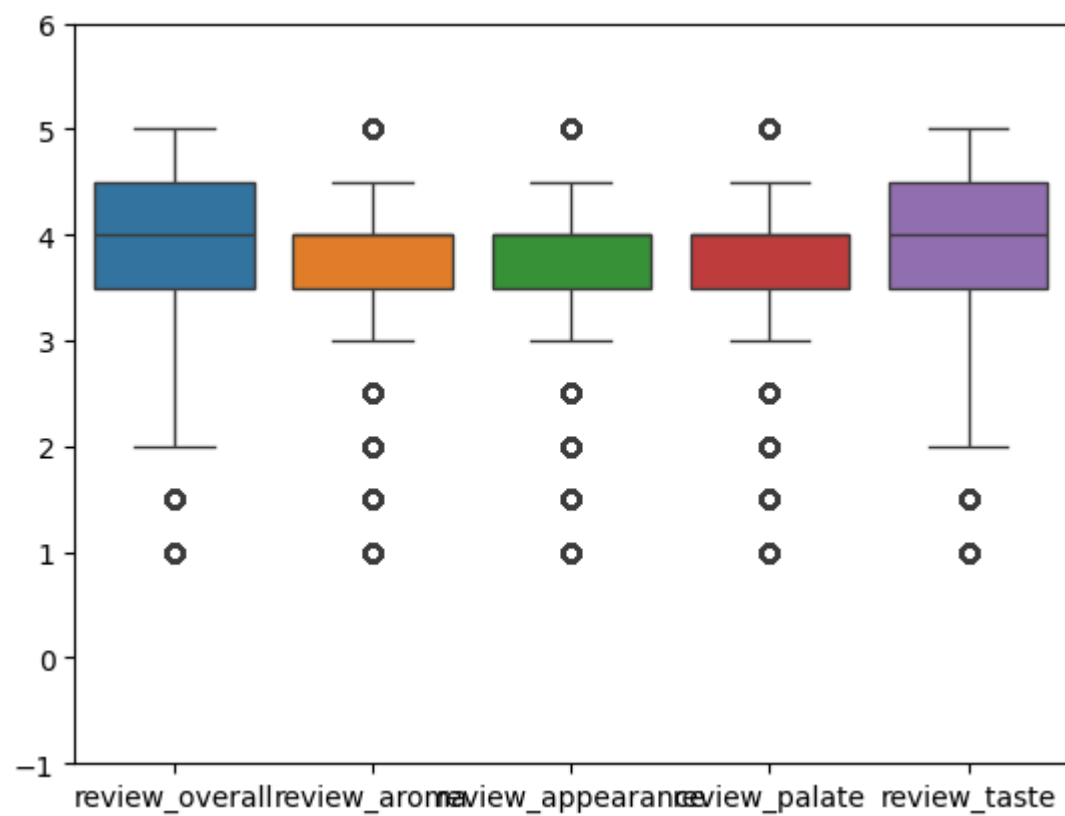


- The rows where review values lie below 1 are removed as reviews should always be above 1

```
In [42]: #beer_data_out = beer_data[beer_data.type==float].drop('beer_abv')
for j in beer_data.columns:
    if beer_data[j].dtype == 'float64':
        beer_data = beer_data[beer_data[j]>=1]
```

```
In [43]: sns.boxplot(data=beer_data.drop(columns=['beer_abv']))
plt.ylim(-1,6)
```

```
Out[43]: (-1.0, 6.0)
```

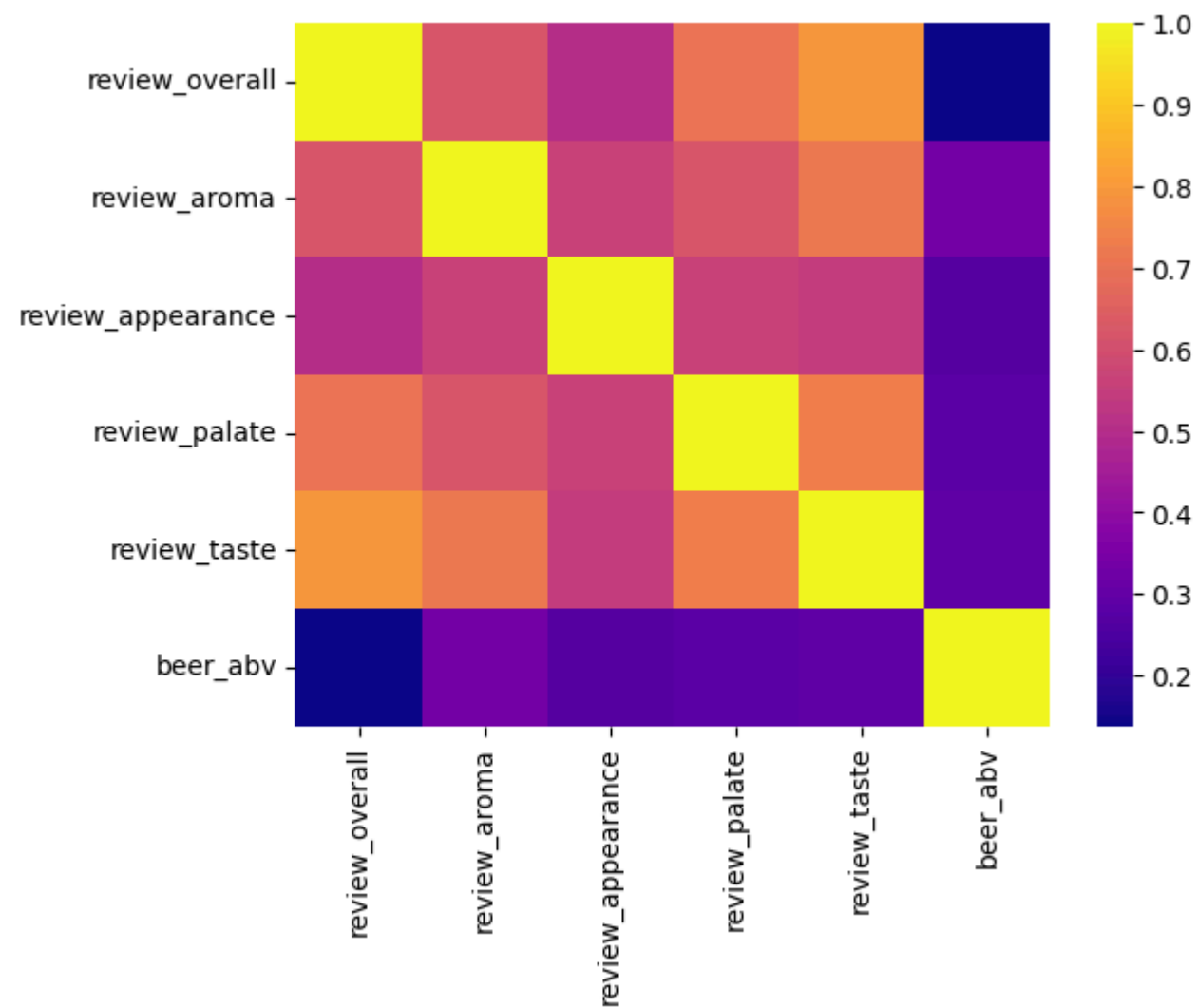
EDA

Correlations

```
In [15]: #Let explore the correlation between the variables
sns.heatmap(beer_data.corr(), cmap='plasma')
```

```
/tmp/ipykernel_229101/2961193758.py:2: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to False. Select only valid columns or specify the value of numeric_only to silence this warning.
  sns.heatmap(beer_data.corr(), cmap='plasma')
```

```
Out[15]: <Axes: >
```



Best beer styles

Insights:

- All the review columns are well correlated with each other whereas a minor corellation can be observed between alcohol percentage and review columns.

Top 10 beer styles

```
In [47]: review_by_style= beer_data.groupby(['beer_style'])['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()
style_counts = beer_data['beer_style'].value_counts()
review_by_style=review_by_style.reset_index()
review_by_style['style_review_count'] = review_by_style['beer_style'].map(style_counts)
review_by_style
```

/tmp/ipykernel_229101/613060892.py:1: FutureWarning: Indexing with multiple keys (implicitly converted to a tuple of keys) will be deprecated, use a list instead.
review_by_style= beer_data.groupby(['beer_style'])['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()

Out[47]:

	beer_style	review_overall	review_aroma	review_taste	review_palate	review_appearance	beer_abv	style_review_count
0	Altbier	3.824415	3.624338	3.745185	3.713519	3.809358	5.829155	7737
1	American Adjunct Lager	3.001546	2.480147	2.683005	2.735322	2.788583	4.861857	30726
2	American Amber / Red Ale	3.780864	3.627064	3.695888	3.656479	3.810159	6.031922	45725
3	American Amber / Red Lager	3.564490	3.213318	3.373333	3.354669	3.527431	4.959722	9296
4	American Barleywine	3.896673	4.019311	4.042588	3.996482	4.036469	10.703947	26721
...	...	...	...	...	...	...	...	...
99	Vienna Lager	3.762461	3.433952	3.603654	3.560181	3.698257	5.020328	8948
100	Weizenbock	4.007919	4.044643	4.077381	3.990699	4.009301	8.135389	9408
101	Wheatwine	3.815244	3.968758	3.976838	3.940749	3.906814	10.662830	3713
102	Winter Warmer	3.703839	3.707107	3.718532	3.670532	3.844621	6.585113	20656
103	Witbier	3.776721	3.640709	3.651978	3.625606	3.686550	5.477745	30126

104 rows × 8 columns

In [48]:

```
#Highest number of reviews
print('Most Popular beer Styles:')
pd.DataFrame(beer_data['beer_style'].value_counts()):10]
```

Most Popular beer Styles:

Out[48]:

	beer_style
	American IPA
	117451
	American Double / Imperial IPA
	85893
	American Pale Ale (APA)
	63426
	Russian Imperial Stout
	54102
	American Double / Imperial Stout
	50679
	American Porter
	50454
	American Amber / Red Ale
	45725
	Belgian Strong Dark Ale
	37733
	Fruit / Vegetable Beer
	33851
	American Strong Ale
	31933

In [49]:

```
#Best mean reviews for beer styles
#review_by_style['number_of_reviews']= beer_data['beer_style'].value_counts(sort='beer_style')
best_style={'Best mean overall review':review_by_style.sort_values(by='review_overall',ascending=False).reset_index().loc[0],
            'Best Taste':review_by_style.sort_values(by='review_taste',ascending=False).reset_index().loc[0],
            'Best Aroma':review_by_style.sort_values(by='review_aroma',ascending=False).reset_index().loc[0],
            'Best palate':review_by_style.sort_values(by='review_palate',ascending=False).reset_index().loc[0],
            'Best Appearance':review_by_style.sort_values(by='review_appearance',ascending=False).reset_index().loc[0]
}
pd.DataFrame(best_style).T
```

Out[49]:

	index	beer_style	review_overall	review_aroma	review_taste	review_palate	review_appearance	beer_abv	style_review_count
Best mean overall review	20	American Wild Ale	4.092915	4.126427	4.14962	4.040259	4.005089	7.713437	17785
Best Taste	41	Eisbock	3.977094	4.156778	4.211603	4.113594	3.964514	11.392852	2663
Best Aroma	11	American Double / Imperial Stout	4.029519	4.160372	4.186951	4.098344	4.163381	10.604766	50679
Best palate	86	Quadrupel (Quad)	4.071614	4.132528	4.210889	4.124952	4.117901	10.461547	18083
Best Appearance	89	Russian Imperial Stout	4.022873	4.076439	4.149449	4.086734	4.209983	9.98665	54102

Scaling

- Scaling is required to put the reviews and beer_abv on the same scale

In [19]:

```
#feature engineering
#review_by_style=review_by_style.reset_index()
X= review_by_style.drop(columns=['beer_style','style_review_count'])
MM= MinMaxScaler()
X_MM = MM.fit_transform(X)
```

/home/jaim/anaconda3/envs/platonium/lib/python3.9/site-packages/sklearn/utils/_array_api.py:472: RuntimeWarning: All-NaN slice encountered
 return xp.asarray(numpy.nanmin(X, axis=axis))
/home/jaim/anaconda3/envs/platonium/lib/python3.9/site-packages/sklearn/utils/_array_api.py:489: RuntimeWarning: All-NaN slice encountered
 return xp.asarray(numpy.nanmax(X, axis=axis))

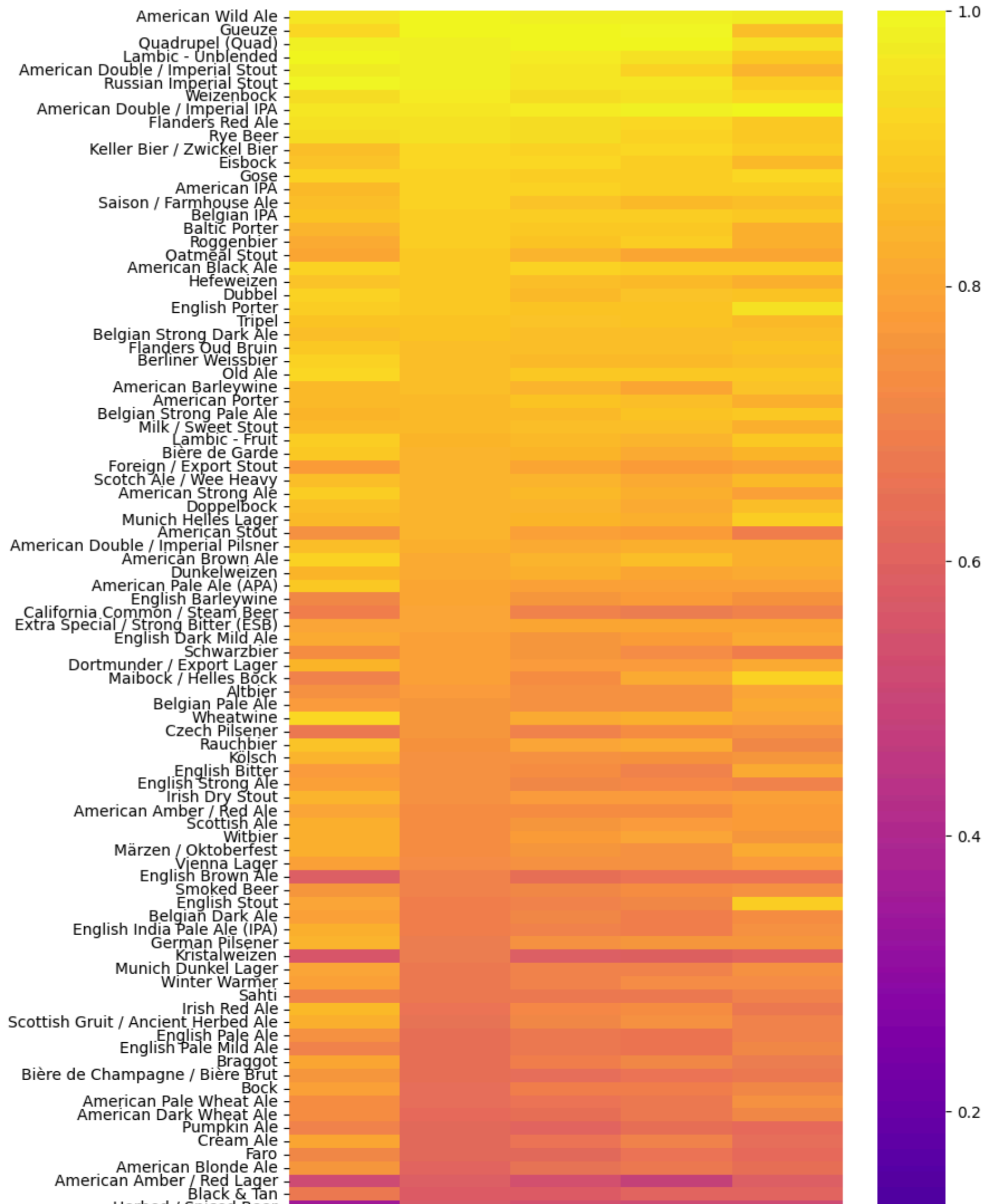
The beer style Heatmap

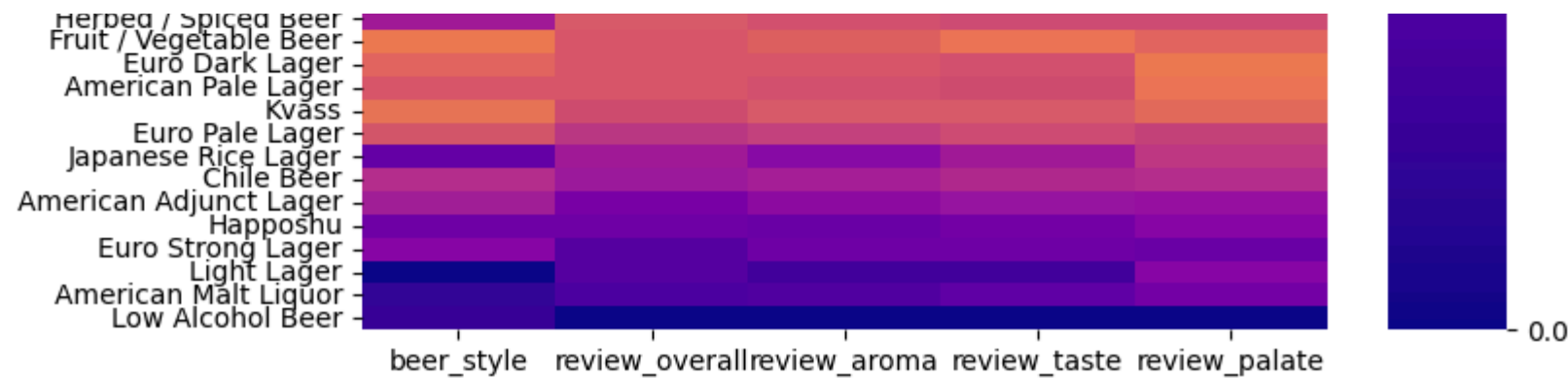
The Heatmap below compares ratings and mean alcohol percentage of all the beer styles.

In [20]:

```
plt.figure(figsize=(8,16))
sns.heatmap(pd.DataFrame(X_MM[:,1:6],columns=review_by_style.columns[1:6]).sort_values(by='review_overall',ascending=False),cmap='plasma',yticklabels=review_by_style.sort_values(by='review_overall',ascending=False).beer_style)
```

Out[20]: <Axes: >





Insights:

- American Wild Ale is the beer style with the highest overall rating whereas Low alcohol beer is the lowest rated beer style.

Best Beers

```
In [21]: review_by_beer=beer_data.groupby(['beer_name'],sort='beer_name')['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()
review_counts = beer_data['beer_name'].value_counts()
review_by_beer=review_by_beer.reset_index()
review_by_beer['beer_review_count'] = review_by_beer['beer_name'].map(review_counts)
print('Most Popular beers:')
display(review_by_beer.sort_values(by='beer_review_count',ascending=False)[:5])
#review_by_beer_data=beer_data.groupby(['beer_name'],sort='beer_name').agg(mean_review_overall=('review_overall','mean'),mean_review_aroma=('review_aroma','mean'),mean_review_taste=('review_taste','mean'),

/tmp/ipykernel_229101/1816600539.py:1: FutureWarning: Indexing with multiple keys (implicitly converted to a tuple of keys) will be deprecated, use a list instead.
review_by_beer=beer_data.groupby(['beer_name'],sort='beer_name')['review_overall','review_aroma','review_taste','review_palate','review_appearance','beer_abv'].mean()
Most Popular beers:
```

	beer_name	review_overall	review_aroma	review_taste	review_palate	review_appearance	beer_abv	beer_review_count
900	90 Minute IPA	4.145338	4.213285	4.325259	4.181140	4.191956	9.000000	3282
26068	India Pale Ale	3.842761	3.758229	3.798019	3.755992	3.856024	6.467234	3129
36774	Old Rasputin Russian Imperial Stout	4.174221	4.198650	4.342173	4.231758	4.373995	9.000000	3111
45693	Sierra Nevada Celebration Ale	4.167391	4.081524	4.188941	4.075175	4.224190	6.800000	2993
52446	Two Hearted Ale	4.329232	4.266067	4.318215	4.137899	4.153324	7.000000	2723

```
In [22]: review_by_beer= review_by_beer[review_by_beer.beer_review_count>20]
best_beers={'Best mean overall review':review_by_beer.sort_values(by='review_overall',ascending=False).reset_index().loc[0],
            'Best Taste':review_by_beer.sort_values(by='review_taste',ascending=False).reset_index().loc[0],
            'Best Aroma':review_by_beer.sort_values(by='review_aroma',ascending=False).reset_index().loc[0],
            'Best palate':review_by_beer.sort_values(by='review_palate',ascending=False).reset_index().loc[0],
            'Best Appearance':review_by_beer.sort_values(by='review_appearance',ascending=False).reset_index().loc[0]
}
pd.DataFrame(best_beers).T
```

Out[22]:

	index	beer_name	review_overall	review_aroma	review_taste	review_palate	review_appearance	beer_abv	beer_review_count
Best mean overall review	41159	Rare D.O.S.	4.848485	4.757576	4.848485	4.80303	4.469697	10.604766	33
Best Taste	31301	M Belgian-Style Barleywine	4.75	4.785714	4.857143	4.803571	4.482143	11.6	28
Best Aroma	31301	M Belgian-Style Barleywine	4.75	4.785714	4.857143	4.803571	4.482143	11.6	28
Best palate	31301	M Belgian-Style Barleywine	4.75	4.785714	4.857143	4.803571	4.482143	11.6	28
Best Appearance	10735	Cantillon Blåbær Lambik	4.628205	4.528846	4.628205	4.49359	4.644231	5.0	156

Best breweries

In [23]:

```
review_by_brew = beer_data.groupby('brewery_name', sort=True).agg(  
    mean_review_overall=('review_overall', 'mean'),  
    mean_review_aroma=('review_aroma', 'mean'),  
    mean_review_taste=('review_taste', 'mean'),  
    mean_review_palate=('review_palate', 'mean'),  
    mean_review_appearance=('review_appearance', 'mean'),  
    beer_abv=('beer_abv', 'mean'),  
    number_of_beers=('beer_name', 'nunique'),  
    number_of_reviews= ('brewery_name', 'count')  
)  
  
display("most popular breweries:", review_by_brew.sort_values(by='number_of_reviews',ascending=False)[:5])
```

'most popular breweries:'

	mean_review_overall	mean_review_aroma	mean_review_taste	mean_review_palate	mean_review_appearance	beer_abv	number_of_beers	number_of_reviews
brewery_name								
Boston Beer Company (Samuel Adams)	3.689435	3.593031	3.647085	3.641303	3.809259	6.406112	133	39433
Dogfish Head Brewery	3.835763	3.916706	3.971082	3.895594	3.943613	9.503279	116	33820
Stone Brewing Co.	4.040084	4.094543	4.150471	4.073404	4.152362	8.251317	134	33043
Sierra Nevada Brewing Co.	4.055942	3.896470	4.028528	3.964962	4.096829	6.908922	140	28726
Bell's Brewery, Inc.	3.972709	3.938088	4.011282	3.965161	3.983395	7.549706	121	25173

In [24]:

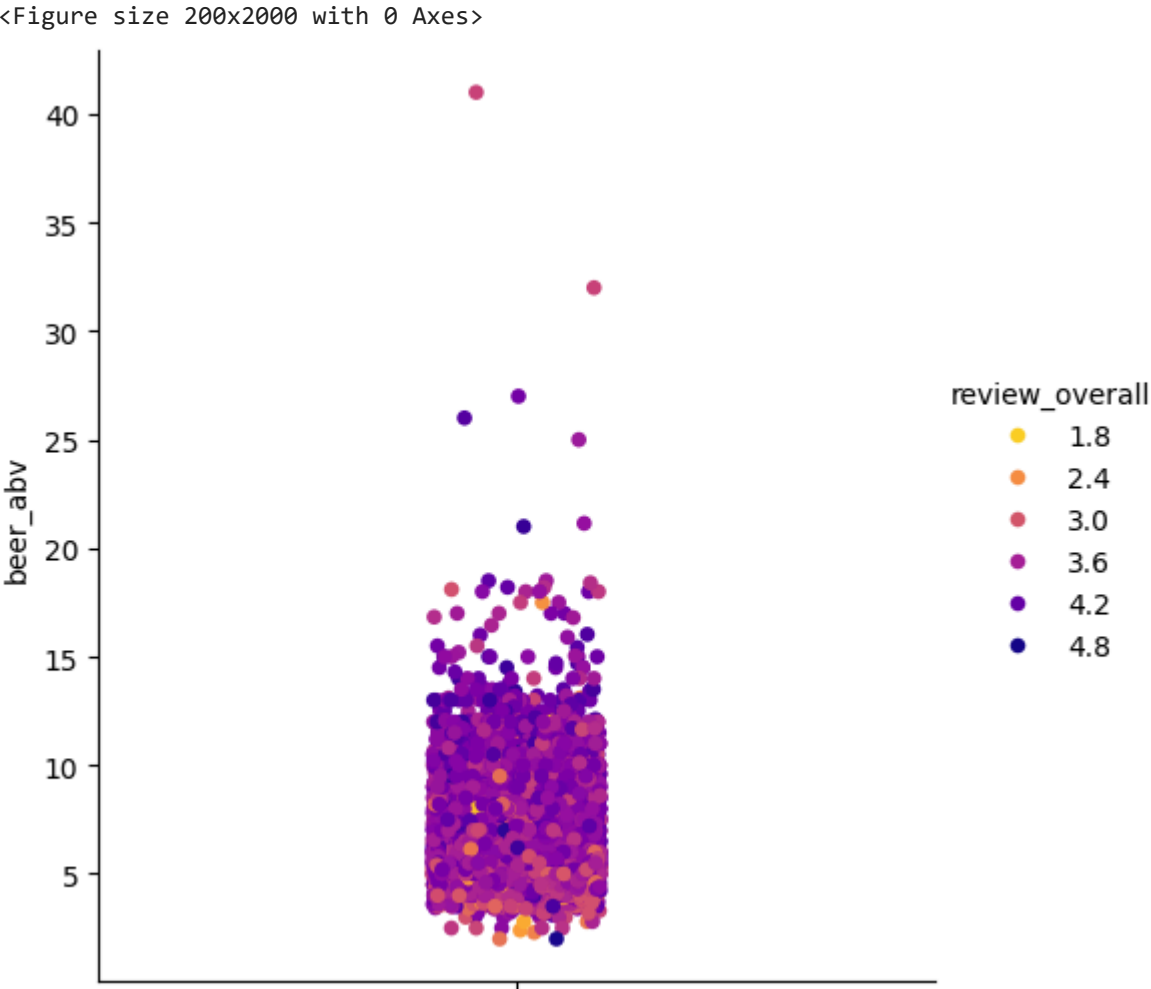
```
review_by_brew= review_by_brew[review_by_brew.number_of_reviews>20]  
  
best_beers={'Best mean overall review':review_by_brew.sort_values(by='mean_review_overall',ascending=False).reset_index().loc[0],  
    'Best Taste':review_by_brew.sort_values(by='mean_review_taste',ascending=False).reset_index().loc[0],  
    'Best Aroma':review_by_brew.sort_values(by='mean_review_aroma',ascending=False).reset_index().loc[0],  
    'Best palate':review_by_brew.sort_values(by='mean_review_palate',ascending=False).reset_index().loc[0],  
    'Best Appearance':review_by_brew.sort_values(by='mean_review_appearance',ascending=False).reset_index().loc[0]  
}  
  
pd.DataFrame(best_beers).T
```

Out[24]:

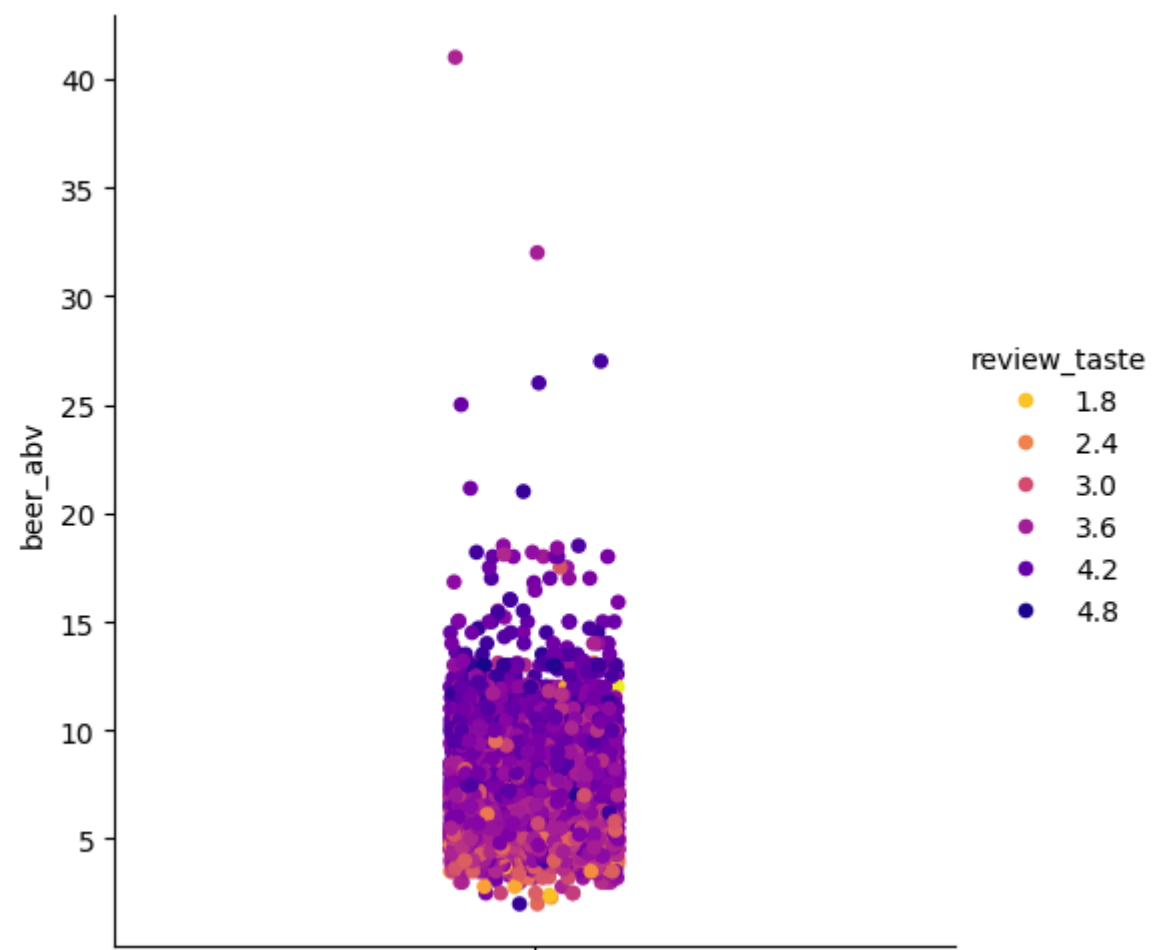
	brewery_name	mean_review_overall	mean_review_aroma	mean_review_taste	mean_review_palate	mean_review_appearance	beer_abv	number_of_beers	number_of_reviews
Best mean overall review	Brauerei Zehendner GmbH	4.607143	4.196429	4.357143	4.375	4.285714	5.583407	3	28
Best Taste	Brouwerij Westvleteren (Sint-Sixtusabdij van W...	4.544996	4.468881	4.58852	4.526493	4.40328	8.808663	5	2378
Best Aroma	The Alchemist	4.581107	4.608779	4.570611	4.388359	4.170802	8.268989	3	524
Best palate	Brouwerij Westvleteren (Sint-Sixtusabdij van W...	4.544996	4.468881	4.58852	4.526493	4.40328	8.808663	5	2378
Best Appearance	Brouwerij Westvleteren (Sint-Sixtusabdij van W...	4.544996	4.468881	4.58852	4.526493	4.40328	8.808663	5	2378

In [25]:

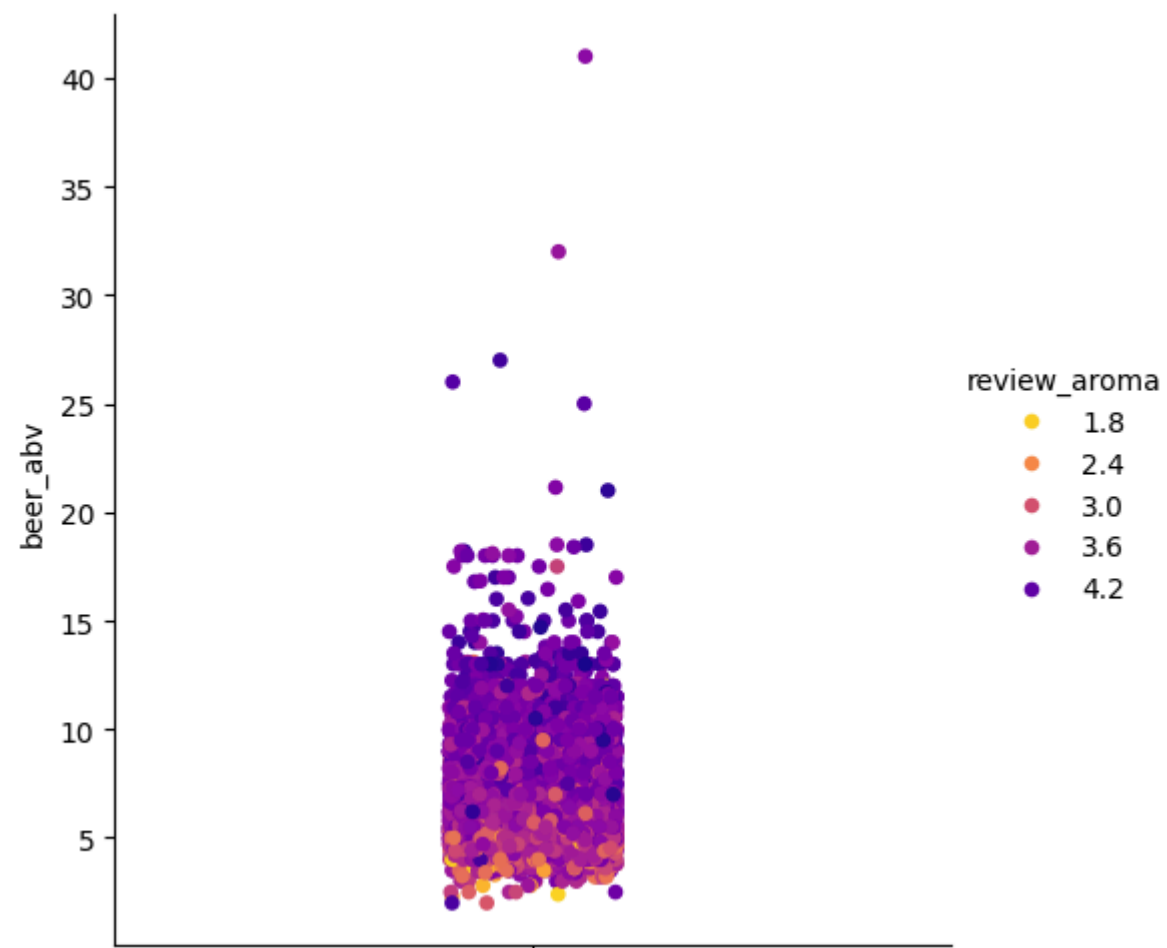
```
coll=['review_overall','review_taste','review_aroma','review_appearance','review_palate']
for col in coll:
    plt.figure(figsize=(2,20))
    sns.catplot(data=review_by_beer,y='beer_abv',hue=col,palette='plasma_r',s=30)
```



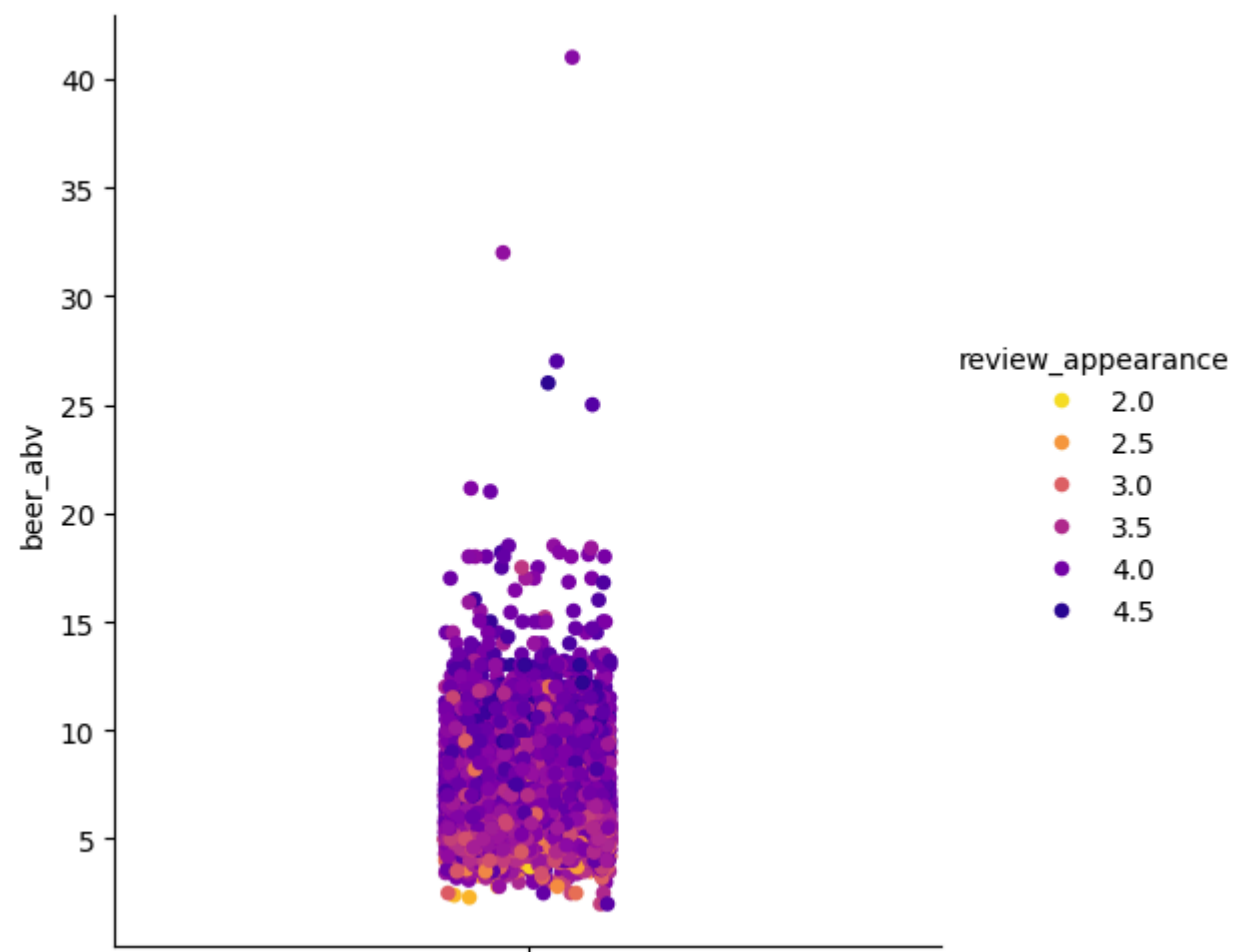
<Figure size 200x2000 with 0 Axes>



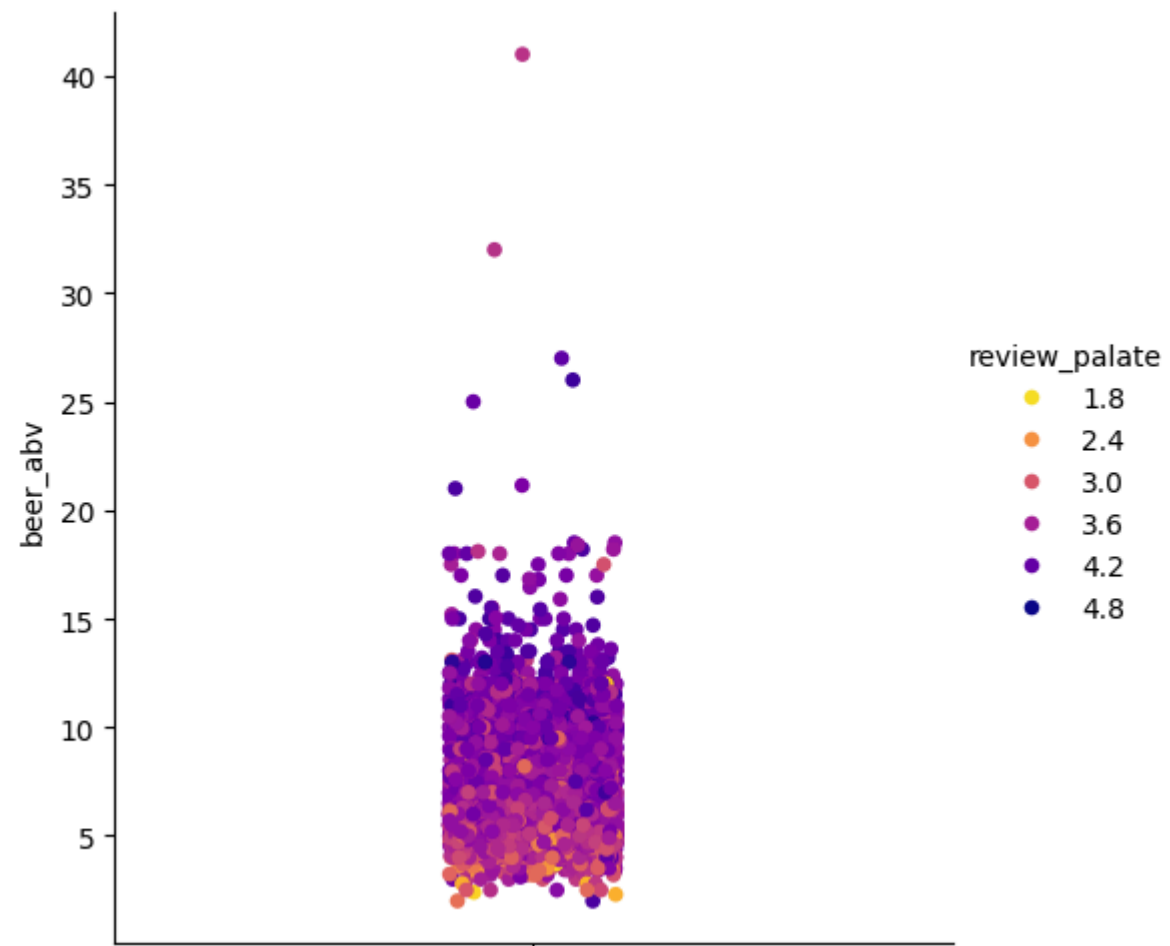
<Figure size 200x2000 with 0 Axes>



<Figure size 200x2000 with 0 Axes>



<Figure size 200x2000 with 0 Axes>

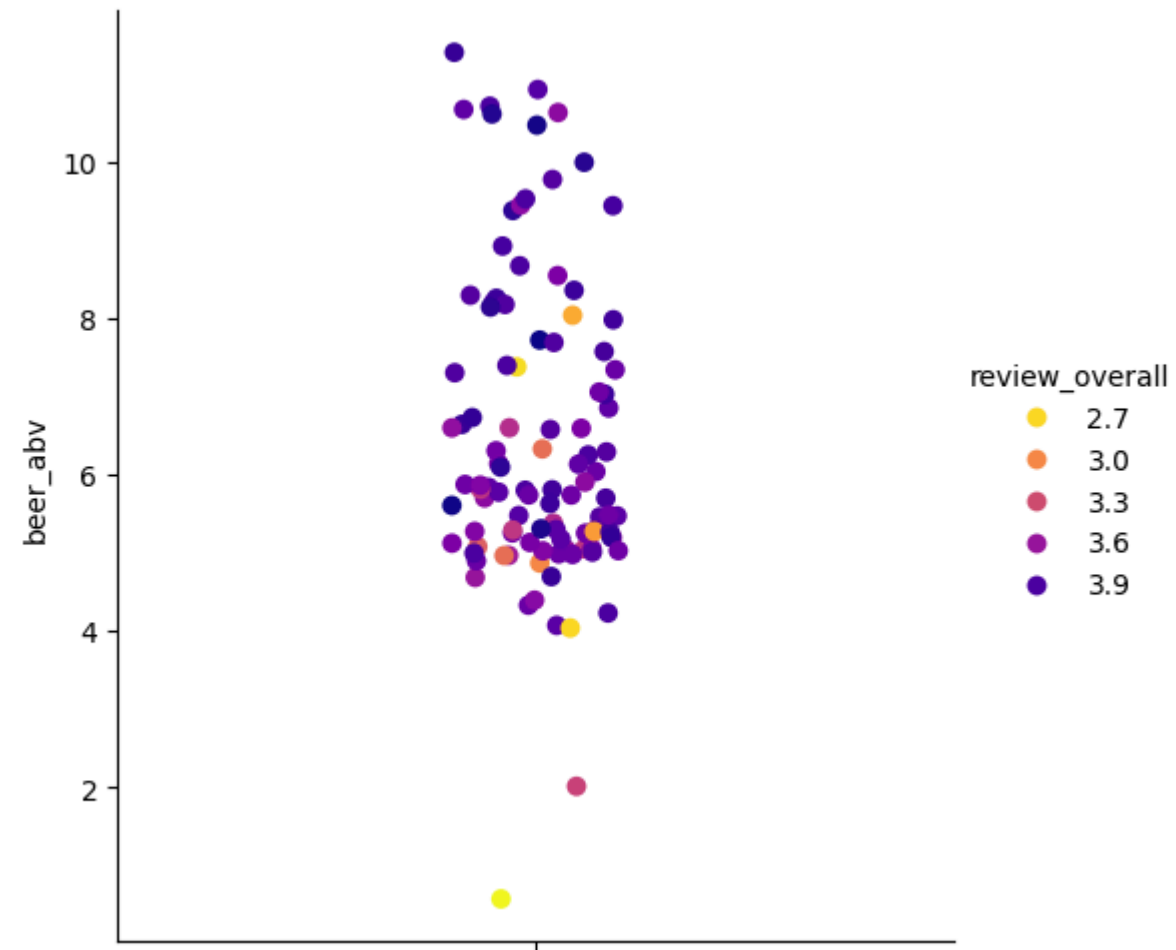


```
In [26]: coll=['review_overall','review_taste','review_aroma','review_appearance','review_palate']  
for col in coll:
```

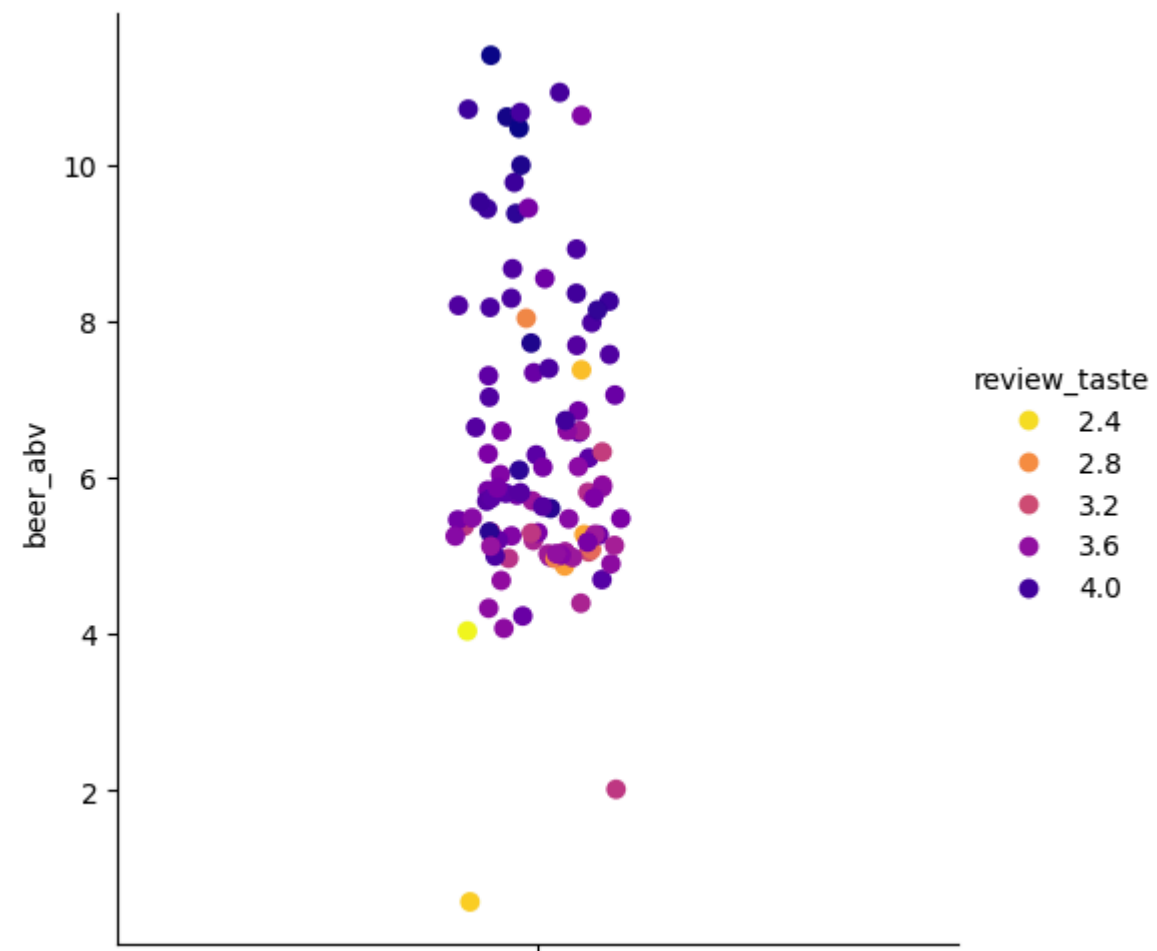


```
plt.figure(figsize=(2,20))
sns.catplot(data=review_by_style,y='beer_abv',hue=col,palette='plasma_r',s=50)
```

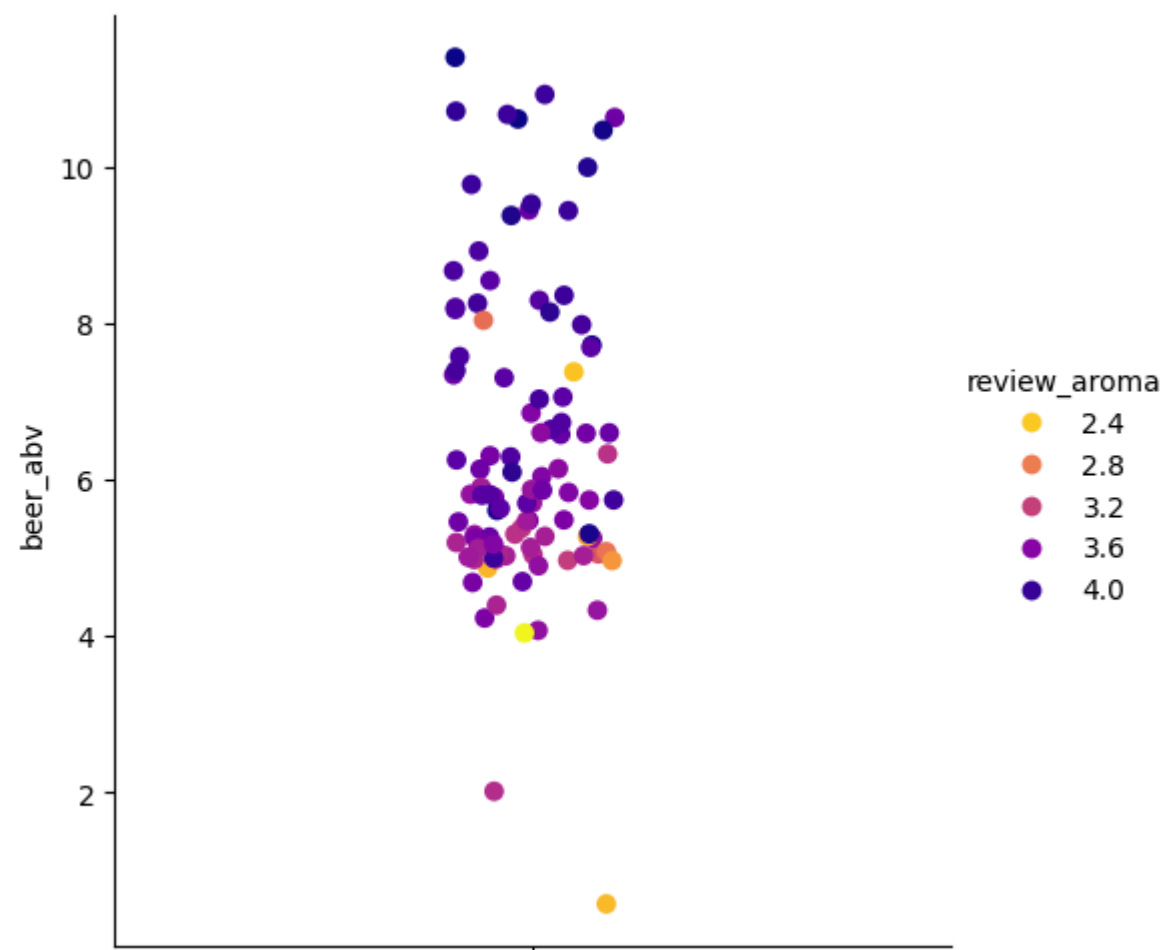
<Figure size 200x2000 with 0 Axes>



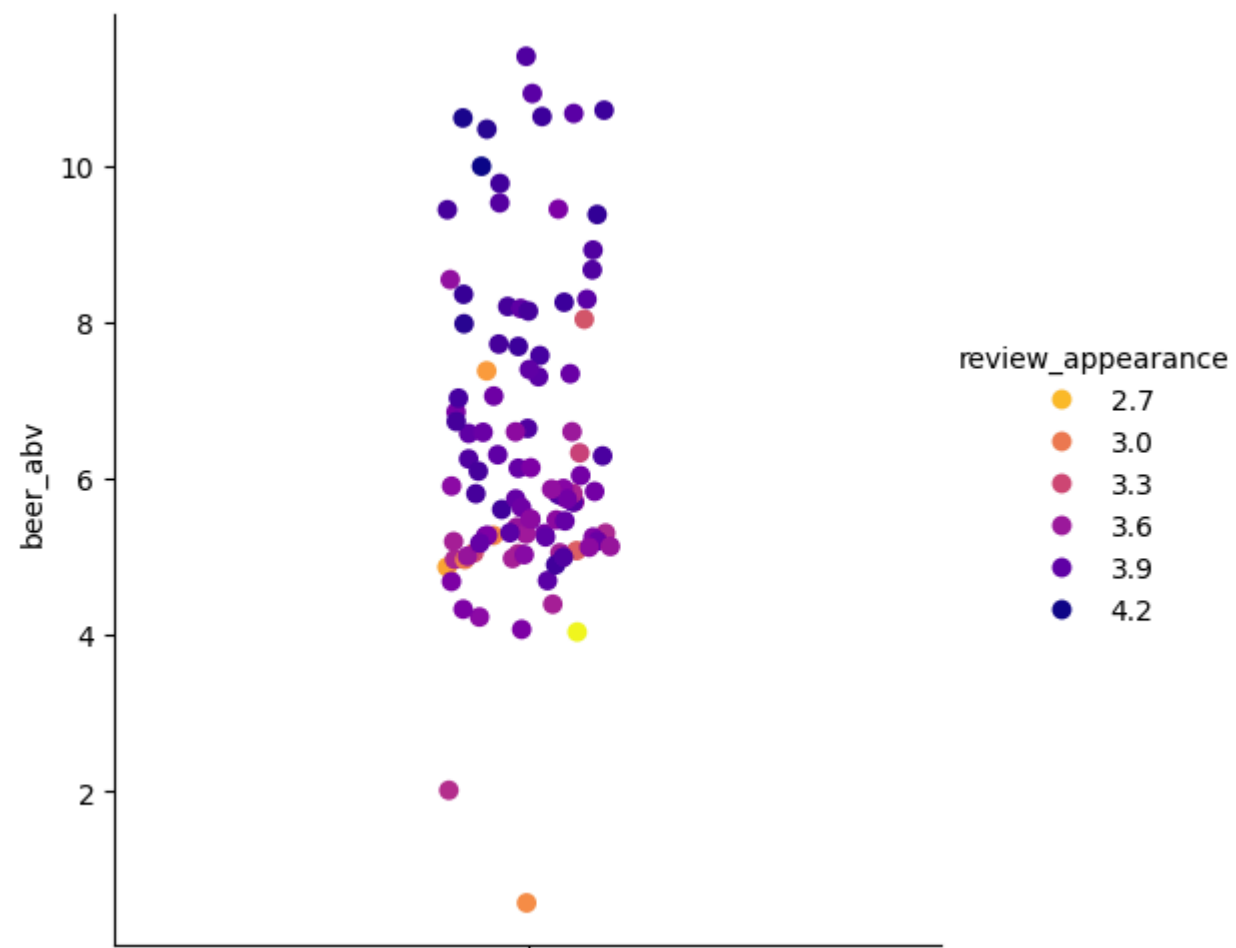
<Figure size 200x2000 with 0 Axes>



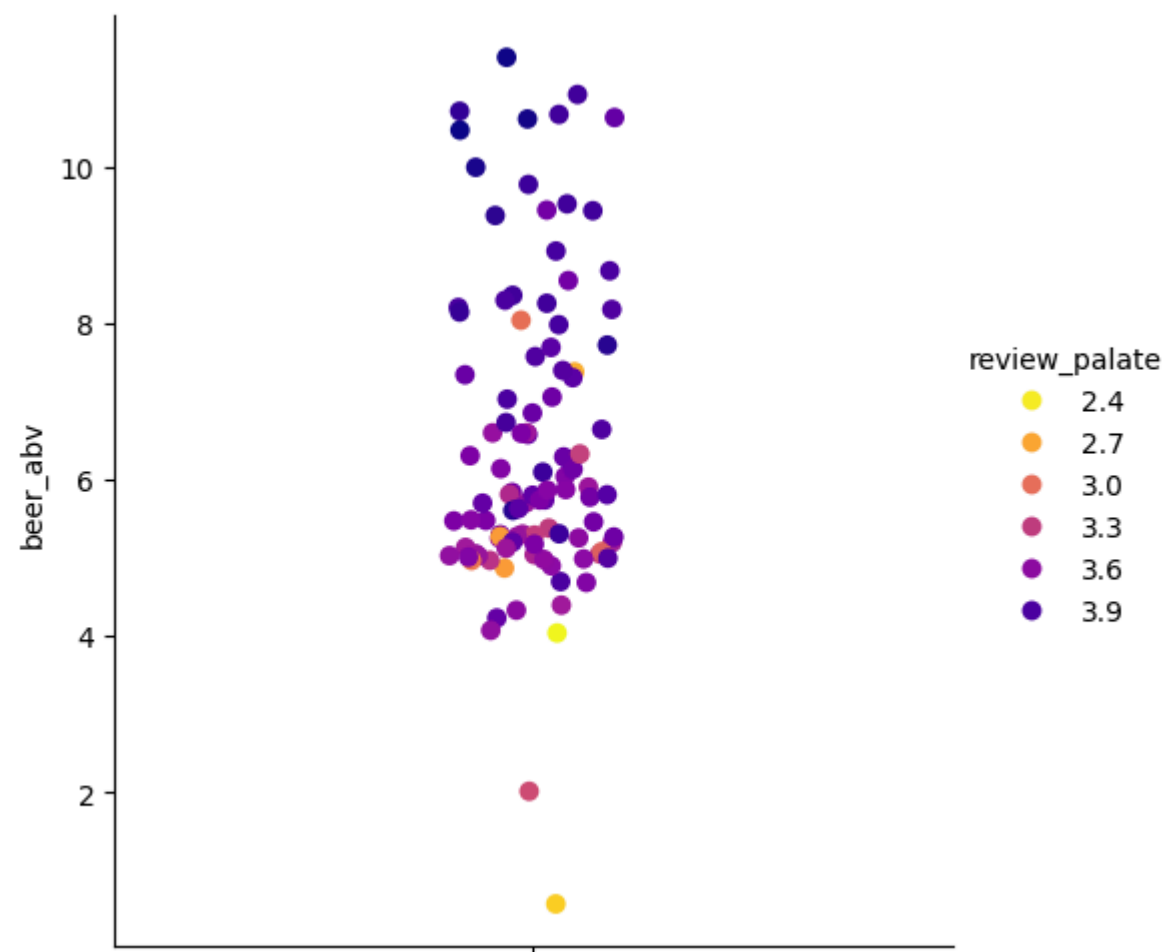
<Figure size 200x2000 with 0 Axes>



<Figure size 200x2000 with 0 Axes>



<Figure size 200x2000 with 0 Axes>



- There are very few low alcohol beers that are highly rated and vice versa

Results

1. Dataset Overview and Preprocessing

- The dataset included comprehensive detailed fields: brewery name, beer style, review aspect scores (aroma, taste, appearance, palate, and overall), and alcohol by volume (ABV).
- Missing ABV values were effectively imputed using the mean ABV for each beer style, and non-informative columns were removed, resulting in a focused and consistent analytical base.

2. Most Popular Beer Styles

- American IPA is the most extensively reviewed beer style, with over 117,000 reviews, indicating its leading popularity. This is followed by:
 - American Double / Imperial IPA: 85,893 reviews.
 - American Pale Ale (APA): 63,426 reviews.
- Other notable styles: Russian Imperial Stout, American Double/Imperial Stout, American Porter.
- **High review volume of American beer suggests an overfitting towards American taste buds as expected due to majority of American reviewers on Beer Advocate**

3. Top-Rated Beer Styles by Review Category

- American wild Ale had the highest mean overall score (3.82), indicating strong quality perception.
- Eisbock led in taste (mean score 4.21).
- American Double / Imperial Stout was top for aroma (mean score 4.19).

- Quadrupel (Quad) led palate (mean score 4.21).
- Russian Imperial Stout topped appearance (mean score 4.15).
- These findings show an appreciation for both classic European and bold American beer styles.

4. **Top-Rated Individual Beers**

- Rare D.O.S. achieved the best mean overall review (4.85).
- M Belgian-Style Barleywine obtained highest marks for taste, aroma, and palate categories.
- Cantillon Blåbær Lambik scored highest for visual appearance.

5. **Leading Breweries**

- By review volume: Boston Beer Company (Samuel Adams) had the most reviews (39,433, across 133 beers), followed by Dogfish Head Brewery and Stone Brewing Co.
- By rating quality: Brauerei Zehendner GmbH had the highest average overall rating (4.61), while niche European breweries like Brouwerij Westvleteren dominated taste, palate, and appearance subcategories.
- The Alchemist led for aroma.

6. **Strength of Beer and Review Scores**

- Beers with higher ABV (above 8%) such as Barleywines, Imperial Stouts, and Quads received higher average review scores for taste and palate, suggesting a preference among seasoned reviewers for strong, complex brews.

In []: